

Calculus Math Notes

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Chapter 1

Limits

Used in defining the important concepts in calculus such as continuity, derivative of a function, the definite integral of a function.

1.1 What are Limits

It's easy to brush off limits due to their inherent simplicity in solving for them, but it's good to know what is happening when you say a limit equals something. This means that as you zoom in on that one area it is *approaching*, the number it equals, despite it potentially never reaching it, still is the answer. The limit is the number you inch closer and closer to while zooming in.

1.1.1 Intuitive Definition

The **limit** of a function describes the behavior of a function near a particular value of x .

When a limit heads towards infinity due to a vertical asymptote this is said to be a **infinite limit**.

1.2 One-Sided Limits

When a limit has a '-' sign it approaches from the *left*.

When a limit has a '+' sign it approaches from the *right*.

1.3 Trigonometric Functions

There are four important limit functions:

1. $\lim_{x \rightarrow c} \sin x = \sin c$

2. $\lim_{x \rightarrow c} \cos x = \cos c$

3. $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

4. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$

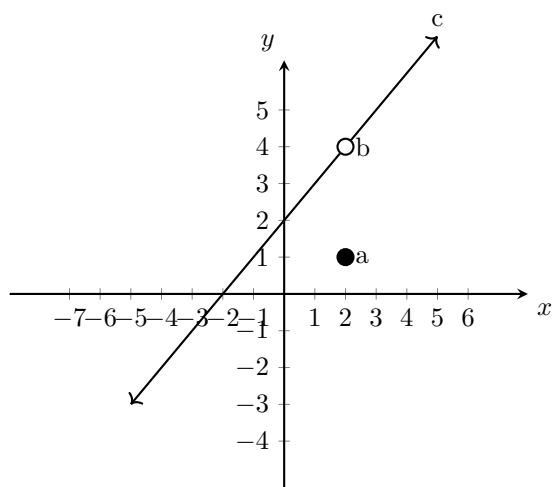
1.4 Continuity

Three conditions must be met for a point to be considered continuous:

1. $f(c)$ exists

2. $\lim_{x \rightarrow c} f(x)$

3. $\lim_{x \rightarrow c} f(x) = f(c)$



In the graph above we can say multiple things:

1. A limit exists at $f(b)$ which is 4.
2. The limit *at* $f(b)$ is 1 aka 'a'.
3. This line has a disconnected continuity.

A test would be to imagine a pencil going along $f(x)$ without having to be picked up – if it can accomplish this then it is continuous.

Chapter 2

Derivatives

2.1 What is a Derivative

It is the instantaneous rate of change which says how much something changes in relation to another. It's the slope of the tangent line at any given point revealing steepness and direction (increasing/decreasing).

2.1.1 Definition

The derivative of a function $y = f(x)$ at point $(x, f(x))$, is defined as:

$$\lim_{\delta x \rightarrow 0} \frac{f(x+\delta x) - f(x)}{\delta x}$$

Example: Using the definition, find the derivative of $f(x) = 5x - 4$

You will want to add in delta into $f(x)$ when you utilize the definition.

$$\begin{aligned} L &= \lim_{\Delta x \rightarrow 0} \frac{(5(x + \Delta x) - 4) - (5x - 4)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{5x + 5\Delta x - 4 - 5x + 4}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{5\Delta x}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} 5 \\ &= 5 \end{aligned}$$

Note that **Differentiability implies continuity, but continuity does not imply differentiability**

2.2 Differentiation Rules

1. If $f(x) = c$ where c is a constant, then $f'(x) = 0$.
2. If $f(x) = c * g(x)$, then $f'(x) = c \cdot g'(x)$.
3. Sum Rule: If $f(x) = g(x) + h(x)$ then $f'(x) = g'(x) + h'(x)$.
4. Difference Rule: If $f(x) = g(x) - h(x)$, then $f'(x) = g'(x) - h'(x)$.
5. Product Rule: If $f(x) = g(x) \cdot h(x)$, then $f'(x) = g'(x)h(x) + g(x)h'(x)$.
6. Quotient Rule: If $f(x) = g(x)/h(x)$, then $f'(x) = \frac{g'(x) \cdot h(x) - h'(x)g(x)}{[h(x)]^2}$

2.3 Trigonometric Functions

The 6 trigonometric functions also have rules:

1. $f(x) = \sin x$, then $f'(x) = \cos x$
2. $f(x) = \cos x$, then $f'(x) = -\sin x$
3. If $f(x) = \tan x$, then $f'(x) = \sec^2 x$
4. If $f(x) = \cot x$, then $f'(x) = -\csc^2 x$
5. If $f(x) = \sec x$, then $f'(x) = \sec x \tan x$
6. If $f(x) = \csc x$, then $f'(x) = -\csc x \cot x$

2.3.1 Inverse Trigonometric Functions

The 6 inverse trigonometric functions also have derivatives:

$$1. \text{ If } f(x) = \sin^{-1} x = \arcsin x \\ f'(x) = \frac{1}{\sqrt{1-x^2}}$$

$$2. \text{ If } f(x) = \cos^{-1} x = \arccos x \\ f'(x) = \frac{-1}{\sqrt{1-x^2}}$$

$$3. \text{ If } f(x) = \tan^{-1} x = \arctan x \\ f'(x) = \frac{1}{1+x^2}$$

$$4. \text{ If } f(x) = \cot^{-1} x = \operatorname{arccot} x \\ f'(x) = \frac{-1}{1+x^2}$$

$$5. \text{ If } f(x) = \sec^{-1} x = \operatorname{arcsec} x \\ f'(x) = \frac{1}{x\sqrt{x^2-1}}$$

$$6. \text{ If } f(x) = \csc^{-1} x = \operatorname{arccsc} x \\ f'(x) = \frac{-1}{\sqrt{1-x^2}}$$

2.4 Chain Rule

The **chain rule** is a technique for finding the derivative of composite functions. With the number of differentiation steps dependent on how many functions there are.

Example 1:

$$f(x) = (g \circ h)(x) = g[h(x)] \text{ then} \\ f'(x) = g'[h(x)] \cdot h'(x)$$

Example 2:

$$r(x) = (m \circ n \circ p)(x) = m\{n[p(x)]\} \\ \text{then } r'(x) = m'\{n[p(x)]\} \cdot n'[p(x)] \cdot p'(x)$$

2.5 Differentiation of Exponential and Logarithmic Functions

Here are their differentiations formulas:

1. If $f(x) = e^x$, then $f'(x) = e^x$.
2. If $f(x) = a^x, a > 0, a \neq 1$, then $f'(x) = (\ln a) \cdot a^x$.
3. If $f(x) = \ln x$, then $f'(x) = \frac{1}{x}$.
4. If $f(x) = \log_a x, a > 0, a \neq 1$, then $f'(x) = \frac{1}{(\ln a) \cdot x}$.

2.6 Implicit Differentiation

Implicit Differentiation allows you to find the derivative of y with respect to x without having to solve the given equation for y .

Example: Find y' when $y = \sin x + \cos y$

$$\begin{aligned}y' &= \cos x - \sin y \cdot y' \\y' + \sin y \cdot y' &= \cos x \\y'(1 + \sin y) &= \cos x \\y' &= \frac{\cos x}{(1 + \sin y)}\end{aligned}$$

2.7 Higher Order Derivative

The degree of derivative depends on how many successive derivatives where taken which are called **higher order derivative**.

Example: Find the second derivative of $f(x) = 20x^3 - 9x^2 + 3x + 2$.

$$\begin{aligned}f'(x) &= 60x^2 - 18x + 3 \\f''(x) &= 120x - 18\end{aligned}$$

Chapter 3

Applications of the Derivative

3.1 Tangent and Normal Lines

The normal line is defined as the line that is perpendicular to the tangent line

3.2 Critical Points

The point $(x, f(x))$ is called a critical point if $f'(x)$ is in the domain of the function and is 0 or doesn't exist.

An important application of critical points is in finding the maxima and minima of a function.

3.3 Extreme Value Theorem

The Extreme Value Theorem helps us guarantee a max and min under certain conditions on an interval.

Extreme Value Theorem.

If a function $f(x)$ is continuous on a closed interval $[a, b]$, then $f(x)$ has a max and min value on $[a, b]$.

3.4 Mean Value Theorem

The **Mean Value Theorem** establishes a relationship between the slope of a tangent line to a curve and a secant line through points on a curve at the endpoints of an interval. The theorem goes:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Mean Value Theorem.

If a function $f(x)$ is continuous on a closed interval $[a,b]$ and differentiable on an open interval (a,b) , then at least one number $c \in (a,b)$.

3.5 Increasing/Decreasing Functions

The derivative may be used to determine whether a function is increasing, decreasing, or an inflection point. If $f'(x) > 0$ then it is increasing but if $f'(x) < 0$ then it is decreasing.

You first find all critical points in the domain value then test the values to the left and right of them to determine if the critical points are positive or negative.

3.5.1 First Derivative Test for Local Extrema

If the derivative switches signs around the critical point, the function is said to have a *local (relative) extremum*.

1. Positive to Negative : *local (relative) maximum*
2. Negative to Positive : *local (relative) minimum*

Determining this is called the **First Derivative Test for Local Extrema**

3.5.2 Second Derivative Test for Local Extrema

This focuses on concavity or the bend the of the line. It is used to find the same the local extremas like with the first test. The main benefit being the this test is quicker but can be inconclusive and relies on being able to derive the second derivative.

You apply it by finding the critical points and testing whether they

are concave up or down.

1. Local Maximum: $f''(c) < 0$ concave down.
2. Local Minimum: $f''(c) > 0$ concave up.
3. Inconclusive: $f''(c) = 0$ tells you nothing.

3.5.3 Concavity and Points of Inflection

If the concavity of a function changes from concave up to down or vice versa around a point then this is a inflection point. You find these by applying the first derivative test.

3.6 Distance, Velocity, Acceleration

1. $f'(x)$ = Velocity of particle
2. $f''(x)$ = Acceleration of particle

3.7 Related Rates

Related rates is just watching how two rates relate given an invisible variable of time.

Picture a balloon. Air goes in. The balloon's radius grows. The volume grows too. You're not asked "what is the radius" or "what is the volume." You're asked: if one is changing this fast, how fast is the other changing right now?

Key:

1. Write the geometric formula.
2. Pretend every variable depends on time.
3. Differentiate both sides with respect to time.
4. Plug in the numbers for the instant you care about.

3.8 Differentials

A differential is a tiny change and they talk about impact. A fast change with a tiny coefficient can matter less than a slow change with a huge one. Differentials make that visible.

Chapter 4

Integration

The Antiderivative, the integral.

4.1 Antiderivative

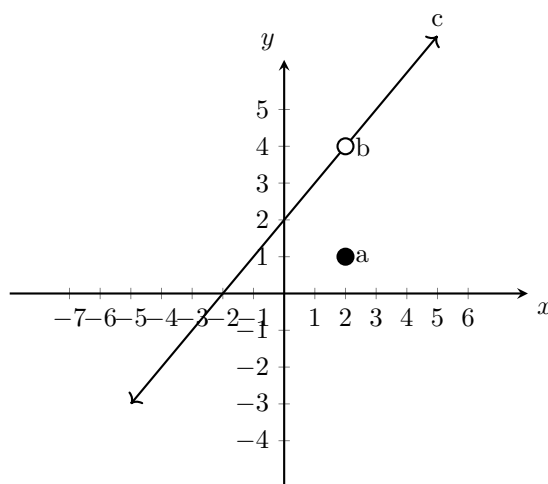
The **Antiderivative** is simply the inverse of the derivative. There are infinitely many antiderivatives for functions. An example would be $F(x) = x^3, G(x) = x^3 + 5, H(x) = x^3 - 2$ being antiderivatives of $f(x) = 3x^2$. That being said they are only differing by a constant – they are only different by their vertical position.

The notation to represent all antiderivatives of a function $f(x)$ is the indefinite integral; symbol is $\int$, where $\int f(x)dx = F(x) + C$. $f(x)$ is called the integrand and the C is called the constant of integration.

4.1.1 Basic Formulas

1. $\int k f(x) dx = k \int f(x) dx$
2. $\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$
3. $\int k dx = kx + C$
4. $\int x^n dx = \frac{x^{n+1}}{n+1} + C$

4.2 Substitution



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